Independent audit of resonant Boussinesq channels: classical charge conservation, heavy instability, and conditional quantum conclusions

Quantum Boussinesq solitons project

4 November 2026

Abstract

(c) Every heavy soliton has an exactly on-shell, opposite-moving light pair, with matching M, Π, P, E and the classical hierarchy. Independent Riccati calculations through H_8 confirm the printed charge decompositions; a Darboux generating function establishes the chord structure without the package's invalid commuting-background argument. The exact Y family also gives a localized growing mode of every interior heavy soliton: instability is already classical. The real one-loop mismatch is reproduced conditionally, not promoted to a physical decay width. Printed threshold-slope and singular-example numbers, and the elementary factor in the Gamma fusion formula, fail independent checks. **(d)** An exact quantum S-matrix, quantum charge selection rules and decay widths remain unestablished.

1 Scope, conventions and reproducibility

Labels have the repository meanings: (a) literature theorem, (b) literature heuristic, (c) independent project derivation/computational evidence, (d) unestablished physical interpretation. "Confirmed" certifies the stated mathematical scope; "qualified" means conditional, restricted or incompletely justified; "refuted" identifies a concrete discrepancy. Each new calculation is in `scripts/audit_resonant_channels.py`. No package code is imported. No git operation, existing-file edit, bulk phase-script run or new renormalisation choice was made.

(c) Read: `AGENTS.md`, `README.md`, the complete target TeX and its 16-page PDF text, all three target scripts, the format of `audit_phase2i.tex`, the ranked next steps, and relevant controlling passages of `errata.tex`, `audit_smx.tex`, `audit_defq.tex`, `audit_det_projected.tex` and `audit_good_renorm.tex`. The last audit matters because even the real one-loop dispersion depends on the finite u^2 subtraction. No external reference, abstract or full text was newly read; no inherited literature attribution is independently authenticated here. All mathematical findings below are labelled (c), rather than claiming an unread literature proof.

(c) We retain

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \quad u = \varphi_x = 2(\log\tau)_{xx}, \quad \pi = \varphi_t, \qquad \Omega = \int d\pi \wedge d\varphi.$$

The physical densities are $M : u$, $\Pi : \pi$, $P : -\pi u$ and $E : (\pi^2 + u^2 + u_x^2)/2 - u^3$. Direct integration agrees with $M = 2k$, $\Pi = -2kv$, $P = 2k^3 v/3$, $E = 2k^3(5 - 4k^2)/15$, $k^2 + v^2 = 1$. For the bad equation the dispersion instead has $k = \sinh\beta$, $v = \epsilon\cosh\beta$, $\omega^2 = k^2 + k^4$; no bad-equation quantum prescription is transferred here. The inherited source dictionary is $u_{\mathrm{CZ}} = -3\varphi$, $\mathcal{L}_{\mathrm{CZ}} = 9\mathcal{L}$. Scattering uses $s = \sigma_{\mathrm{vac}}$, $q_D = -s/(2\sqrt{3})$; the spectral parameter is $\eta\sigma_{\mathrm{vac}}$. These must not be interchanged. The target uses only an abstract directed-edge holonomy, so it does not reproduce the refuted signed scattering substitution in its computations.

2 Verdict table: all labelled claim groups

Repeated claims in the abstract and conclusion are covered by their numbered source proposition and the rows below. The equation names are the source's LaTeX labels. Every row has independent checks in `audit_resonant_channels.py` unless explicitly identified as an unestablished physical claim (d).

Claim	Verdict	Evidence and precise scope
C01. One-body and KP dictionary; cross-ratio (Sec. 1, Prop. 2.1)	Confirmed (c)	Bilinear polynomial, all four directions, physical charge identities.
C02. Three resonance curves (Prop. 2.1.1)	Qualified	Curves confirmed (c); printed $\ell > 0$ fails for left-moving overtaking. Equal-soliton diagonal excluded.
C03. KP triangle and projections (Prop. 2.1.2; abstract)	Qualified	Main chord formulas confirmed (c). First abstract formula for α_{13} is false; static $\ell = 0$ is an endpoint exception.
C04. Positive Y solution, splitting/merging (Prop. 2.2.1)	Qualified	Y solution confirmed (c). Literal heavy-plus-light fusion into the other light is refuted by positive mass; shared-root projections survive.
C05. Areas and measure (Prop. 2.2.2)	Confirmed (c)	Independent slice integrals: singular fractions 2/9, 2/3; two-body measure zero does not restrict heavy one-body measure.
C06. $H_1, \ldots, H_8$ and potentials (Thm. 3.1.1–2)	Confirmed (c)	New spectral series plus direct Riccati/beta integrations; printed H_7, H_8 correct.
C07. All-order proof and spin gap (Thm. 3.1.3)	Qualified	Conclusion established by a different formal Darboux proof (c); printed commuting-background argument invalid. Spin-gap assertion confirmed.
C08. Classical telescoping, topology, Π (Prop. 3.2.1)	Confirmed (c)	All-order generator and M, Π, P, E integrated through Y collision. Original script integrates only P, E.
C09. Mode shifts and $\Delta F = 0.38879244\ldots$ (Prop. 3.2.2)	Qualified	Number and mode telescoping confirmed (c) by regulated root-transmission traces; finite subtraction and heavy-saddle determinant conditional (d).
C10. Shifted quantum fusion curve (Remark 3.3)	Qualified	Local E, P curve persists (c); simultaneous quantum charges and physical dispersion unresolved (d).
C11. $h \log h$, delays and Taylor logarithm (Prop. 4.1)	Qualified	Generic classical coefficients confirmed (c), away from cusps; physical resummation/frame interpretation conditional (d).
C12. Light concavity and soliton $1 \to 2$ protection (Thm. 4.3.1)	Confirmed (c)	Symbolic velocity differentiation; concavity plus positive mass. Subadditivity requires a defined total light momentum.
C13. Every-heavy embedding and Jacobian (Thm. 4.3.2)	Qualified	Exact all-heavy charge matching confirmed (c); $1/(\sqrt{3}k)$ is one ordered head-on contribution, not the unrestricted displayed integral.
C14. On-shell slope value (`onshell_slope`)	Refuted (c)	Symbolic formula correct; printed $+0.241078\ldots$ is actually $-0.2664334283\ldots$.
C15. WKB deficit (Prop. 4.4)	Qualified	Damped-sheet algebra confirmed (c); opposite sheet amplifies. Neither a physical-sheet selection nor inelastic probability derived (d).
C16. Singular numeric example (Prop. 4.4)	Refuted (c)	All three printed A, $\lvert\Delta\ell\rvert$, $\lvert S\rvert^2$ values disagree.

Claim	Verdict	Evidence and precise scope
C17. Classical and phonon fusion (Prop. 4.5.1–2)	Confirmed (c)	Hirota factors/phases and independently reconstructed phonon transmission telescope; absolute phase constants still conditional (d).
C18. Gamma fusion obstruction (Prop. 4.5.3)	Qualified	Implemented candidate fails, with quoted phase (c); printed elementary-factor exponent is reversed. Physical bootstrap premise unproved (d).
C19. Target-momentum remainder (Prop. 4.5.3)	Qualified	0.8755292547... logarithmic part confirmed (c); not the full Darboux Taylor remainder or a physical no-go.
C20. Four regimes, width, 2×2 block and chronological failure (Sec. 5; remarks)	Qualified (d)	Classical domain classification survives (c). No quantum vertex, spectral normalization, complete channels or quantum charges computed.
C21. Singular two-phase breathers	Qualified	Elementary bound $A_{\mathrm{br}} < 1/4$ confirms singularity within this ansatz (c); does not exclude quantum bound states (d).
C22. Script completeness and timings (Sec. 6)	Qualified	All three scripts pass in 21.87 s (c); several advertised checks are absent or only assumed identities.

3 Hierarchy charges: a separate derivation

Riccati and scattering generating functions (C06–C08). **(c)** Put $L_0 = D^3 - D/4$ and solve $\lambda^3 - \lambda/4 = z^3$ with $\lambda = z + O(z^{-1})$. For the eigenfunction logarithmic derivative $w = \psi_x/\psi$,

$$w_{xx} + 3ww_x + w^3 + qw + r = z^3, \qquad w = z + \sum_{n\geq 1} w_n z^{-n}. \tag{1}$$

The integrated formal trace identity is $H_n = -n\int (w_n - w_{n,\mathrm{vac}})\,dx$; the discarded local terms are total x derivatives. It can be obtained by differentiating the formal eigenvalue problem with respect to z and cyclically moving D under the integrated residue. In particular $w_1 = -q/3$, $w_2 = (q_x - r)/3$, giving $H_1 = \int (q - q_0)/3$ and $H_2 = 2\int r/3$ directly. Our recursive scalar ODE computation is independent of the package's fractional pseudodifferential multiplication.

(c) For $g = e^{-p_a x} + e^{-p_b x}$, $f = g_x/g$ and $B = D - f$, $BL_0 = LB$ with

$$q = -\tfrac{1}{4} + 3f_x, \qquad r = 3ff_x + 3f_{xx}, \qquad \psi = (\lambda - f)e^{\lambda x}. \tag{2}$$

The vanishing constant coefficient is the derivative of $f_{xx} + 3ff_x + f^3 - f/4 =$ constant; all exponentials have the same L_0 eigenvalue. The endpoint scattering identity gives

$$\int (w - \lambda)\,dx = \log\frac{\lambda + p_b}{\lambda + p_a} = -\sum_{j\geq 1}\frac{\Delta_j}{j\lambda^j}, \qquad \Delta_j = (-p_b)^j - (-p_a)^j, \tag{3}$$

$$H_n = n[z^{-n}]\sum_{j\geq 1}\frac{\Delta_j}{j\lambda(z)^j}. \tag{4}$$

Every coefficient is a polynomial endpoint difference. Thus the formal local hierarchy has chord additivity to all orders, not just to order eight. The common vacuum spectrum is essential; this is not a binomial expansion of noncommuting differential operators.

(c) Independent series extraction in `audit_resonant_channels.py` gives precisely

$$\begin{aligned} &H_1 = \Delta_1, \quad H_2 = \Delta_2, \quad H_3 = H_6 = 0,\\ &H_4 = \Delta_4 - \tfrac{1}{3}\Delta_2, \qquad H_5 = \Delta_5 - \tfrac{5}{72}\Delta_1,\\ &H_7 = \Delta_7 - \tfrac{7}{12}\Delta_5 + \tfrac{7}{324}\Delta_1, \qquad H_8 = \Delta_8 - \tfrac{7}{36}\Delta_4 + \tfrac{11}{324}\Delta_2. \end{aligned}$$

All printed potentials for M, Π, P, E, H_7, H_8 have zero exact remainder modulo $p_a^2+p_ap_b+p_b^2-1/4$. The pure-k^3 ansatz in $\mathrm{span}\{\Delta_1, \Delta_5\}$ gives an inconsistent exact linear system. Independently, direct scalar recursion through w_8 on $u = 2k^2y/(1+y)^2$, $y = e^{kx}$, reduces the integrals to

$$\int_0^\infty \frac{y^j}{(1+y)^N}\,dy = \frac{j!(N-j-2)!}{(N-1)!}.$$

All eight charges agree exactly at $(k, v) = (1/2, \sqrt{3}/2)$ and $(97/103, -20\sqrt{3}/103)$, including a heavy left-mover. These are exact rational/algebraic evaluations, not fitted coefficients.

(c) The target's proposed proof by shifting $\widetilde{L}$ by $-D/4$ is not valid: $[D, \widetilde{L}] = \widetilde{q}_x D + \widetilde{r}_x$ is nonzero on a soliton. Its vague operator logarithm is not a derivation of the claimed zero-background trace formula. Equation (4) supplies the missing argument within the formal scalar hierarchy. No claim about arbitrary quantum/nonlocal charges follows from this formal trace theorem.

Across the actual collision. **(c)** Use the minus-sign Y solution, $g = \sum_i \exp(-q_i x + \sqrt{3}q_i^2 t)$, so the same Darboux transformation applies with three terms. At every finite t, $f(-\infty) = -q_1$, $f(+\infty) = -q_3$. Therefore the entire integrated generating function remains $\log[(\lambda+q_3)/(\lambda+q_1)]$ through the collision. Our checker integrates its spatial density at $t = -5, 1, 7$ and two finite spectral parameters; it does not merely subtract endpoint formulas from themselves. Separate spatial quadratures of all four physical densities at these times match the heavy values. This includes M and Π, unlike the original script, whose actual quadrature loop selects only density indices 2 and 3 (P, E). Hierarchy additivity at collision was not directly tested by that original quadrature.

4 Heavy-soliton kinematics and classical instability

Exact on-shell map, not an isolated heavy parameter (C12–C14). **(c)** For a right-moving heavy soliton $a \in (\pi/3, \pi/2)$, set

$$\alpha_1 = 2\pi/3 - a, \quad \epsilon_1 = +1, \qquad \alpha_2 = a - \pi/3, \quad \epsilon_2 = -1.$$

Then $\alpha_1+\alpha_2 = \pi/3$ and the two chords concatenate into the heavy chord. Reversing all velocities covers the left-moving branch. All four physical charges and the entire formal hierarchy agree exactly. The curve in the two-particle parameter plane has measure zero, but its projection onto the heavy parameter covers the whole open heavy branch. These are distinct measure statements.

(c) No listed topological charge forbids this classical process: the potential endpoints are fixed along the exact solution, M is continuous rather than an integer kink number, and Π is explicitly conserved. The integrated spectral generator also telescopes, so the usual local spectral hierarchy supplies no additional obstruction. A smooth exact trajectory is stronger evidence than matching four asymptotic numbers. **(d)** This does not enumerate all possible quantum superselection rules or compute an anomalous/nonlocal quantum charge. The defensible quantum conclusion is compatibility with the checked classical charges, not a theorem that no hidden quantum selection rule exists.

(c) Using velocity as independent coordinate gives

$$\frac{dE}{dP} = v, \quad \frac{dP}{dv} = \frac{2}{3}\sqrt{1-v^2}(1-4v^2), \quad \frac{d^2E_{\mathrm{light}}}{dP^2} = -\frac{3}{2\sin 3\alpha} < 0.$$

Concavity and monotonicity imply the stated strict inequality when $|P_1+P_2| \le P_{\max}$. If the sum lies outside this interval the right-hand light energy is undefined, so the abstract's unrestricted wording needs this restriction. Mass positivity excludes a heavy daughter in a light-soliton $1 \to 2$ decay. This is soliton-sector kinematic protection, not a general proof of quantum elastic factorisation with arbitrary radiation channels.

(c) At fixed heavy momentum, the derivative of the head-on pair energy with respect to its right-moving momentum is $v_1 - v_2 = \sqrt{3}k_3 > 0$. Hence the ordered head-on delta-function contribution is exactly

$$J_{+-} = \frac{1}{\sqrt{3}k_3}. \tag{5}$$

It is finite and nonzero in the interior, with the strict range $(1/\sqrt{3}, 2/3)$; the printed closed lower endpoint requires the excluded static limit. The displayed unrestricted integral in the package needs a channel restriction (and a convention for identical particle counting). If both orderings are integrated, both roots contribute. There can also be co-moving E, P roots near the cusp; our independent solver finds one at heavy $a = 1.1$. Its mass fails to match, so it does not undermine the charge-conserving head-on result, but it refutes interpreting the displayed integral as an unrestricted total phase space. Its widths are $k_1 = 0.07962397547781387719\ldots$, $k_2 = 0.83585099579192889532\ldots$, with mass mismatch $M_1 + M_2 - M_3 = 0.04853522241661486512\ldots$.

(c) The printed formula $\Delta E'(0) = -(v_2 - v_3)P_2'(\alpha_2)$ is correct. At $\alpha_1 = 0.62$ the signed value is

$$\Delta E'(0) = -0.26643342833433332432785323923 16\ldots, \tag{6}$$

not $+0.24107867527655644$. We differentiate the energy mismatch using an independent velocity inversion of the heavy momentum. The sub-threshold statement $\Delta E(h) < 0$ for $h > 0$ survives; the printed positive slope would have contradicted it. Globally in the light-light square, $\partial_{\alpha_2}\Delta E = (v_2 - v_3)P_2'(\alpha_2) > 0$, since $v_2 < -1/2$ and $|v_3| < 1/2$. The unique zero is the displayed head-on resonance, so the lower triangle is below it. Independent sample checks cover both sides.

An explicit classical unstable mode, absent from the package. (c) In the plus-sign Y family insert a small coefficient $c_2 = \varepsilon$. In the frame $\xi = x - v_3 t$, after removing a harmless exponential,

$$\tau \;\sim\; 2\cosh(k_3\xi/2) + \varepsilon e^{-\sqrt{3}v_3\xi/2}e^{\gamma t}, \qquad \gamma = \sqrt{3}k_1k_2 = \sqrt{3}(1/4 - v_3^2) > 0. \tag{7}$$

Consequently $u = u_0 + \varepsilon e^{\gamma t}\chi + O(\varepsilon^2)$ with

$$u_0 = \frac{k_3^2}{2}\operatorname{sech}^2(k_3\xi/2), \qquad \chi = 2\partial_\xi^2\left[\frac{e^{-\sqrt{3}v_3\xi/2}}{2\cosh(k_3\xi/2)}\right]. \tag{8}$$

Both tails decay precisely because $k_3 > \sqrt{3}|v_3|$, equivalent to $|v_3| < 1/2$. Differentiating the exact family proves, and 80-digit numerical spatial differentiation independently verifies,

$$(\gamma - v_3\partial_\xi)^2\chi = \chi'' - \chi'''' - 6(u_0\chi)''.$$

The associated potential perturbation also decays; its M and Π variations are zero boundary terms. Thus the heavy soliton already has a localized exponentially growing classical mode. **(d)** A real continuum mode trace on this saddle is not by itself its physical quantum energy; discrete unstable modes and the determinant contour/domain must be addressed. The existence of this classical mode does not fix a quantum width or its normalization.

5 One-loop mismatch and direction of the shifted resonance

Independent regulated computation (C09–C10). (c) We reconstruct the phonon transmission from the soliton chord and complex phonon roots

$$z_\pm(q) = \frac{\operatorname{sgn}(q)\sqrt{1+q^2} \pm i\sqrt{3}q}{2\sqrt{3}}, \qquad A_{\mathrm{ph}}(q) = \frac{(p_a - z_+)(p_b - z_-)}{(p_a - z_-)(p_b - z_+)}.$$

We differentiate its logarithm analytically and integrate the regulated energy and momentum traces, including the inherited equal-mode endpoint and tadpole:

$$e_\Lambda = \frac{1}{4\pi}\int_{-\Lambda}^{\Lambda} |q|\sqrt{1+q^2}\,\theta'(q)\,dq - B_\Lambda + \frac{3k}{\pi}(\sqrt{1+\Lambda^2}-1), \tag{9}$$

$$p_\Lambda = \frac{1}{4\pi}\int_{-\Lambda}^{\Lambda} q\theta'(q)\,dq, \quad B_\Lambda = \frac{\Lambda\sqrt{1+\Lambda^2}}{4\pi}[\theta(\Lambda)-\theta(-\Lambda)]. \tag{10}$$

Endpoint arguments are chosen to tend to zero at large $|q|$. No package angular-log charge routine or saved result is used. With $q = \sinh t$, independent cutoffs and a free intercept in $1/\Lambda, 1/\Lambda^3$, the real trace reproduces

$$\Delta F = F_{12} + F_{23} - F_{13} = 0.38879244070035460274\ldots, \qquad F = e - vp, \tag{11}$$

while $\Delta e = \Delta p = 0$. This is conditional on that prescribed trace and finite subtraction. The defect holonomy identity $\sum_i \rho_i \mathcal{O}_i = \eta(\mathcal{O}_{12} + \mathcal{O}_{23} - \mathcal{O}_{13})$ is valid orientation bookkeeping, not a new physical absorption test. It is not a newly selected physical renormalisation.

What the positive sign means. **(c)** At the old classical daughter momenta, keeping all three physical momenta fixed, the corrected daughter energy exceeds the heavy energy by $\hbar\Delta F > 0$. Thus that particular old pair is off shell. It does not follow that the entire decay closes: the classical root is inside the head-on continuum and has a nonzero derivative. At fixed physical heavy P_3, write $P_1 \mapsto P_1 + \hbar\delta p$, $P_2 \mapsto P_2 - \hbar\delta p$. Then

$$\delta p = -\frac{\Delta F}{v_1 - v_2} < 0. \tag{12}$$

For sufficiently small $\hbar$ at any fixed interior heavy parameter, the right-moving daughter momentum decreases and the left-moving daughter momentum becomes less negative. At $\phi = 0.23$, $\delta p = -0.23450418388533775558\ldots$. Both daughter widths decrease locally. The equality moves toward $\alpha_1 + \alpha_2 < \pi/3$, the regular side, in these fixed-physical-momentum coordinates. This is a shifted on-shell locus, not a continuum-edge threshold opening/closing. The statement is nonuniform near the heavy cusp, where one daughter becomes soft.

(c) A finite term $\hbar\lambda u^2$, explicitly left unselected by the controlling good-renormalisation audit, changes the mismatch by

$$\Delta F_\lambda = \Delta F_0 - 2\lambda k_1 k_2 k_3. \tag{13}$$

This follows from $\int u^2 dx = 2k^3/3$ and $k_3 = k_1 + k_2$, and is independently checked. Therefore even the quoted numerical sign is a prescription-dependent real-trace statement. **(d)** Simultaneous conservation of the physical quantum M, Π, H_n, and a physical complex energy of the unstable saddle, cannot be inferred by shifting two momenta in an E, P equation alone. The package correctly acknowledges this higher-charge gap in its remark but overstates instability/elasticity as settled in parts of the conclusion.

6 Resonance phases, formal deficits and fusion

Classical geometry and transverse limits (C01–C05, C11). **(c)** Factoring the pairwise Hirota dispersion gives

$$\mathcal{P}(q_i - q_j, \sqrt{3}(q_i^2 - q_j^2)) = 4(q_i - q_j)^2(q_i^2 + q_i q_j + q_j^2 - 1/4).$$

It proves the positive Y solution and the three shared-root loci. The correct lines are $R_{\rm ho}^+$: $\alpha_1 + \alpha_2 = \pi/3$, $R_{\rm ho}^-$: $|\alpha_1 - \alpha_2| = \pi/3$, and $R_{\rm ov}$: $\alpha_1 + \alpha_2 = 2\pi/3$. No overtaking pole occurs in the open square. The printed $\ell > 0$ on overtaking is true only for right-movers; left-movers have

$\ell < 0$. The first abstract expression $\alpha_{13} = \alpha_{12} + \alpha_{23} + \pi/3$ would give $2\pi/3$, outside the declared range, since the two light angles sum to $\pi/3$. The main-text piecewise heavy angle is correct. At $\phi = \pi/6$ the cubic is nondegenerate but the heavy chord is static, $k = 1$, an excluded endpoint of the stated open one-body parametrisation.

(c) For generic noncusp points, differentiating the independently computed Hirota phase twice gives $\lim_{h\to 0^+} h\delta''(h) = \frac{2}{9}\sigma c_\ell$ with the three tabulated coefficients. We use $h = 10^{-25}$ and an independent derivative extraction, not the package's two-point subtraction. The fixed-P logarithmic coefficient is nonzero at the three sampled points. The singular expansion is not uniform when h is comparable to $\hbar$; interpreting it as a resummed physical branch displacement requires the missing physical frame/determinant.

Damping is a sheet-dependent candidate calculation (C15–C16). (c) The two continuations $\log A = \log|A| \pm i\pi$ give reciprocal squared moduli. On the damped branch only,

$$|e^{i\delta/\hbar}|^2 = e^{-4\pi|\ell_1-\ell_2|/(9\hbar)}.$$

Calling that branch "physical" because it is bounded is an assumption, not a derivation of scattering boundary conditions. A deficient candidate alone does not calculate the channels into which probability flows. Away from the exact on-shell curve, $2 \to 1$ is not energy conserving, so it cannot simply be assigned the entire open-domain deficit. The correct numbers for the printed example $(0.50, 0.65)$, directions $(+,-)$, $\hbar = 0.1$, are

$$\begin{aligned} A &= -3.78262233323924739654273624 79\ldots, \\ |\ell_1 - \ell_2| &= 0.29944362968358401836763860 7696\ldots, \\ |S|^2 &= 0.01528288338741825938604932718 41\ldots. \end{aligned}$$

All three quoted package numbers fail; its script asserts only $A < 0$ and $0 < |S|^2 < 1$, never those numbers. **(d)** Identifying the complement with a physical inelastic probability or proving failure of elastic quantum scattering would require a controlled amplitude.

Fusion survives classically, but the Gamma candidate does not (C17–C19). (c) Independent Hirota factors satisfy $A(3,12)A(3,23) = A(3,13)$; the topological term telescopes using both M and Π. The full real classical phases therefore add. The independently constructed complex phonon transmission itself telescopes pointwise, a stronger test than subtracting angular phase derivatives. With the same integration and reference prescription this proves additivity of the bare integral linear in the target phase derivative. It does not set absolute one-loop phase constants, precisely as the earlier phase audits warn.

(c) Write $R(\eta) = \Gamma(1 + i\eta)/\Gamma(1 - i\eta)$. The implemented Gamma candidate gives the fusion ratio

$$\left[R(\eta)\left(\frac{e^2}{\eta^2}\right)^{+i\eta}\right]^{-\tau(3,13)}, \tag{14}$$

with a *plus* exponent inside the brackets. The package prints a minus exponent. The plus sign cancels the leading Stirling phase and tends to $\exp[-i\pi\tau\,\mathrm{sgn}(\eta)/2]$. Fresh log-Gamma evaluations give the quoted $1.5158110455271881045\ldots$ phase at $\hbar = 0.1$ and approach $+\pi/2$ at 10^{-4} and 10^{-8}; the literal printed formula instead has phase $2.88667602677875534274\ldots$ at $\hbar = 0.1$. Thus the obstruction for this candidate survives, while its displayed identity is refuted. The earlier candidate audit does not identify this candidate as the physical S-matrix.

(c) Using the independently integrated mode momenta gives the target logarithmic remainder $0.8755292547543045976\ldots$. Its nonvanishing is a valid statement about the displayed expression. However the full Darboux shift also contains $\gamma_j M_i$, so calling this logarithmic piece the entire fixed-physical-momentum fusion mismatch is incomplete. In the same mode-charge

prescription, retaining that term gives 3.58464625500292908012… at the quoted point, rather than 0.8755292547…. Moreover the single-soliton momentum choice, two-body frame and finite subtraction remain conditional. **(d)** The proposed chronological failure and a universal physical fusion no-go therefore remain conjectural.

(c) The good two-phase breather bound follows directly from $A_{\mathrm{br}} = (1 - 3\cot^2 a)/(1 + 3\coth^2 b) < 1/4$. The tau polynomial cannot remain positive when its interference minimum is reached with $A < 1$; this is an ansatz-specific classical singularity. **(d)** It supplies neither a quantum bound-state exclusion nor an exact S-matrix pole theorem.

Quantum claims that remain guesses (C20). **(d)** The displayed decay-width normalization needs normalized states and a defined vertex. The proposed 2×2 block assumes a complete two-channel space, while an unstable heavy state is not automatically an asymptotic one-particle channel. One cannot simultaneously infer its instability and use it as an exact stable external state without specifying a resonance formalism. Neither kinematic protection nor a scalar Yang–Baxter identity proves factorisation. The four regimes are useful classical geometry, but "must augment by $2 \leftrightarrow 1$" throughout $A < 0$ and "only these sectors can be elastic" go beyond the checked results. The unresolved quantum vertex and higher-charge corrections are substantive missing work.

7 Verification coverage, numerical budgets and failures recorded

Original scripts. **(c)** Byte-identical copies were executed under new `/tmp/audit_resonant_channels_pkg_*` names. Their only output writes follow `__file__`, so their JSON files also stayed in `/tmp`; tracked caches were untouched. Measured wall times were:

Target script suffix	Seconds	Exit / final marker
`conservation.py`	16.388	0 / present
`kinematics.py`	3.709	0 / present
`quantum.py`	1.773	0 / present

All print `ALL CHECKS PASSED`; the total is 21.870 s on this run, consistent with the claimed under-25-second run, not a portable speed bound.

(c) The hierarchy integrator is substantive independent-of-target code within the package; it is not a fitted formula compared with itself. Conversely:

- The symbolic telescoping loop converts stringified expressions with `sympify` without preserving the original real-symbol assumptions. Its p can differ from the substitution symbol, so these particular substitutions can yield identically zero charges before the check. Even without that bug, cancellation of a supplied F with itself would check bookkeeping rather than derivation. The earlier exact residue integration does supply substantive independent evidence.

- The claimed direct collision tests of M, Π and all H_n are absent; only P, E are spatially integrated. The all-order proof is asserted in prose.

- The phase-space Jacobian is not numerically integrated or explicitly asserted in the quantum checker. Area fractions are saved as strings rather than verified by area integration. Bare phonon derivatives are sampled at four points; the advertised continuum fusion integral is not integrated.

- The unitarity test inserts the claimed positive imaginary part and checks an exponential is between zero and one; it selects no physical sheet. The printed slope and singular example are not compared with computed values. The Gamma comparison uses the corrected plus sign, not the printed minus.

- The one-loop mismatch test asserts only a nonzero discrepancy, not the 20-digit printed number; the independently reconstructed trace here supplies that missing numerical cross-check. No quantum matrix element is tested.

(c) No evidence of tolerance weakening in the available final scripts was found. Their own failures list explicitly admits an initially fabricated trailing-digit target, an incorrect early coefficient guess, insufficient asymptotic time, and root-inversion/sign mistakes. Those admissions are not erased by current passing tests. Editing history cannot be reconstructed from these snapshots.

Independent error accounting. **(c)** Exact symbolic identities have no mesh/fit error. Riccati beta integrals are exact at the two stated profiles; the generic generating-function identities are polynomial identities. Y quadratures use central moments, different panels and times from the package, and 45/65-digit arithmetic; finite-spectral-parameter generator integrals use 80 digits. Their differences are numerical diagnostics, not rigorous interval error bounds. The root-trace calculation separates (i) quadrature precision/panel refinement, (ii) finite cutoff variation, and (iii) two- versus three-cutoff extrapolation spread; no leading target coefficient or mismatch is fed into the fit. These budgets certify conditional integrals, not the uncomputed determinant or a physical renormalisation uncertainty. Asymptotic $h \log h$ extraction uses a declared 10^{-21} tolerance at $h = 10^{-25}$ rather than claiming all 80 digits.

(c) The measured diagnostics are:

Quantity / independent variation	Value
ΔF, $\Lambda = \sinh 18$	0.38879243375721738313
ΔF, $\Lambda = \sinh 24$	0.38879244068314428624
ΔF, $\Lambda = \sinh 30$	0.38879244070031194263
Free three-cutoff intercept	0.38879244070035460274
Two-/three-cutoff intercept spread	6.10×10^{-35}
Trace quadrature method/precision/panel spread	1.66×10^{-47}
Y physical-charge 45-/65-digit spread	8.55×10^{-47}

The trace mesh check compares 80-digit tanh–sinh on t panels $0, 1, 3, 6, 18$ with 60-digit Gauss–Legendre on $0, 1/2, 2, 5, 10, 18$. The finite-cutoff discrepancy at the finest cutoff is about 4.27×10^{-14}, much larger than quadrature error; it is removed by extrapolation, not silently included in a high-precision equality. The two-/three-cutoff spread is model sensitivity, not a statistical confidence interval or an all-orders tail bound. No empirical fit is used for H_n.

Failures recorded.

1. Package-claim discrepancies: left-moving overtaking ℓ sign; first abstract heavy-angle formula; invalid commuting-background proof; unrestricted phase-space formula; slope sign and magnitude; three singular example numbers; Gamma elementary-factor sign; literal heavy-plus-light fusion labels. Refuted claims are encoded as assertions that the discrepancies exist.

2. The first independent symbolic integration attempt grew too slowly with rational (k, v) carrying irrational coefficient denominators and was interrupted. We changed exact test profiles so k and $v/\sqrt{3}$ are rational, retaining both a light and a heavy case and all eight charges. The generic symbolic generating-function identity was retained, not weakened.

3. An initial audit check assumed that merely reversing the printed slope sign would repair it. It failed with a $0.025354753\ldots$ magnitude discrepancy. We replaced that false target with an independently differentiated analytic expression and explicitly assert the printed magnitude discrepancy.

4. The first PDF build reported an overfull temporary-path line. Explicit filename break points repaired it; all pages were rendered again.

Measured independent execution. **(c)** The final complete checker ran in 171.773 seconds and printed `ALL CHECKS PASSED`. Earlier complete successful versions took 141.592 and 180.752 seconds; the final run includes the additional printed-table, phase-space, breather and literal-fusion discrepancy assertions. All three original script copies also printed the marker. Assertions encode refutations as detected discrepancies, not false equalities; no failed tolerance was relaxed. The final PDF was built twice, without overfull-box warnings, and all ten pages were rendered and visually inspected.

Deliverables and references. The independent checker, this TeX and its twice-built PDF are the only new repository files. No existing status or inventory file was edited, as required by the task. Temporary build and inspection files have the same audit prefix. No bibliography is invented: cited local audit filenames identify the exact scope of inherited qualifications, and no external full-text reading is claimed. **(d)** The remaining scientific task is a quantum treatment of the unstable heavy saddle and its coupled channels with physical higher charges, not further refinement of already exact classical charge matching.